ASSIGNMENT-14.4

NAME: K.NAGA CHAITANYA
HALLTICKET:2303A51292
BATCH:05

#TASK 1:
PROMPT:
Generate a responsive homepage using HTML and CSS. The page should include a header with navigation links, a main content section, and a footer. Use CSS media queries to ensure the layout adapts to mobile, tablet, and desktop screen sizes. Keep the design clean and modern.

CODE:
**Index.html code:**

index.html

index.html > html > body > section.features > div.feature-item

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Startup Homepage</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <header>
        <h1>Startup </h1>
        <nav>
            <ul>
                <li><a href="#">Home</a></li>
                <li><a href="#">About</a></li>
                <li><a href="#">Services</a></li>
                <li><a href="#">Contact</a></li>
            </ul>
        </nav>
    </header>

    <section class="hero">
        <h2>Welcome to Our Startup</h2>
        <p>Innovating the future with cutting-edge technology.</p>
        <a href="#" class="button">Get Started</a>
    </section>

    <section class="features">
        <div class="feature-item">
            <h3>Feature 1</h3>
            <p></p>
        </div>
        <div class="feature-item">
            <h3>Feature 2</h3>
            <p></p>
        </div>
        <div class="feature-item">
            <h3>Feature 3</h3>
            <p></p>
        </div>
    </section>

    <footer>
        <p>&copy; 2023 Startup Name. All rights reserved.</p>
        <div class="social-links">
            <a href="#">Facebook</a>
            <a href="#">Twitter</a>
            <a href="#">LinkedIn</a>
        </div>
    </footer>
</body>
</html>
```

Style.css code:

style.css > ...

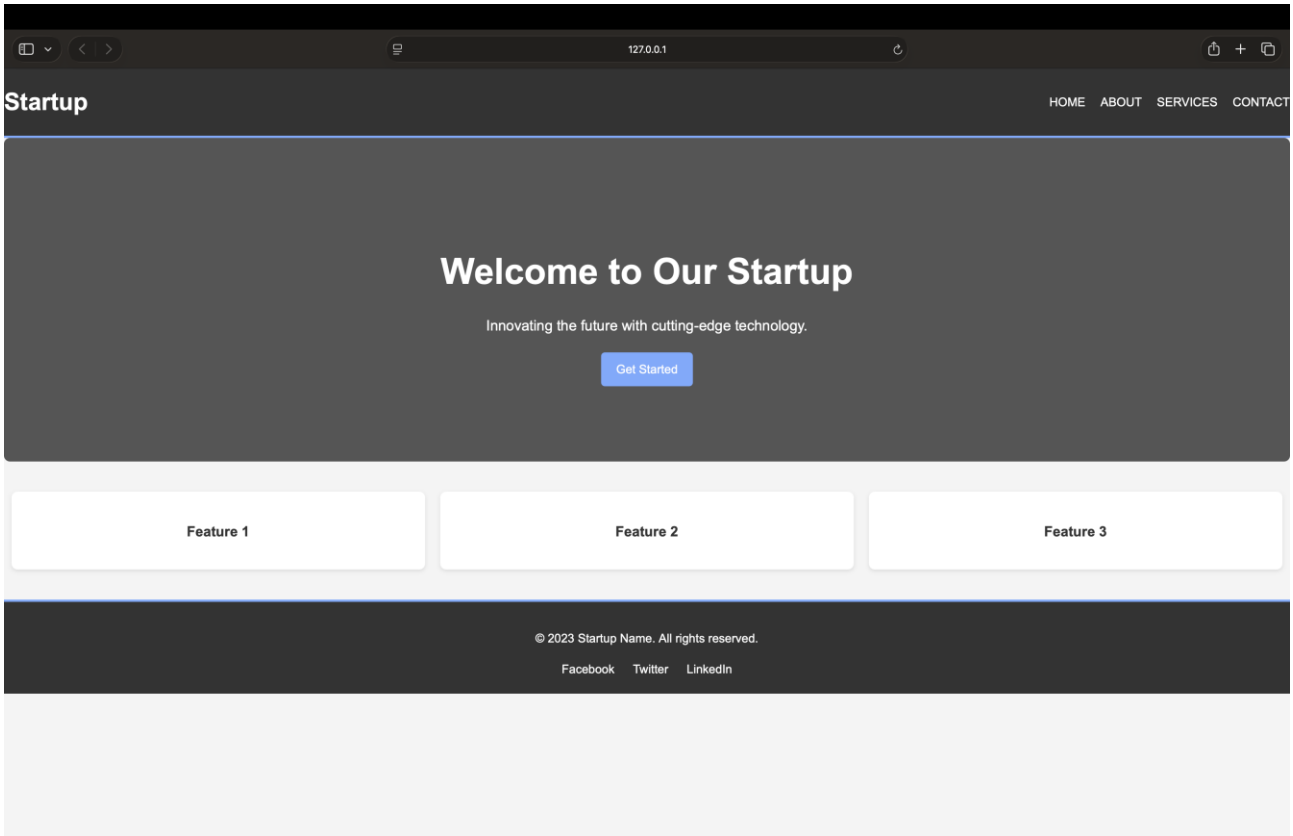
```css
/* Basic Reset & Body Styles */
body {
    margin: 0;
    font-family: Arial, sans-serif;
    line-height: 1.6;
    background-color: #f4f4f4;
    color: #333;
}

.container {
    width: 90%;
    max-width: 1200px;
    margin: auto;
    overflow: hidden;
    padding: 0 20px;
}

/* Header Styles */
header {
    background: #333;
    color: #fff;
    padding: 1rem 0;
    border-bottom: #77aaff 3px solid;
    display: flex;
    justify-content: space-between;
    align-items: center;
}

header h1 {
    margin: 0;
    padding: 0;
    flex-grow: 1;
}

header nav ul {
    padding: 0;
    list-style: none;
    display: flex;
}

header nav ul li {
    margin-left: 20px;
}

header nav a {
    color: #fff;
    text-decoration: none;
    text-transform: uppercase;
    font-size: 16px;
}

header nav a:hover {
    color: #77aaff;
}

/* Hero Section */
.hero {
    background: #555 url('https://via.placeholder.com/1500x500') no-repeat center center/cover;
    color: #fff;
    text-align: center;
    padding: 100px 20px;
    margin-bottom: 30px;
    border-radius: 8px;
}
```

style.css > ...

```
.hero h2 {
    font-size: 3em;
    margin-bottom: 10px;
}

.hero p {
    font-size: 1.2em;
    margin-bottom: 20px;
}

.button {
    display: inline-block;
    color: #fff;
    background: #77aaff;
    padding: 10px 20px;
    text-decoration: none;
    border-radius: 5px;
    transition: background 0.3s ease;
}

.button:hover {
    background: #5588dd;
}

/* Features Section */
.features {
    display: flex;
    justify-content: space-around;
    flex-wrap: wrap;
    margin-bottom: 30px;
}

.feature-item {
    background: #fff;
    padding: 20px;
    margin: 10px;
    border-radius: 8px;
    box-shadow: 0 2px 5px rgba(0,0,0,0.1);
    flex: 1;
    min-width: 280px;
    text-align: center;
}

.feature-item h3 {
    color: #333;
    margin-bottom: 10px;
}

/* Footer Styles */
footer {
    background: #333;
    color: #fff;
    text-align: center;
    padding: 20px 0;
    margin-top: 30px;
    border-top: #77aaff 3px solid;
}

footer .social-links a {
    color: #fff;
    margin: 0 10px;
    text-decoration: none;
}
```

```
# style.css > ...
footer .social-links a:hover {
    color: #77aaff;
}
/* Media Queries for Responsiveness */
/* Tablets (smaller than 768px) */
@media (max-width: 768px) {
    header {
        flex-direction: column;
        text-align: center;
    }

    header nav ul {
        margin-top: 10px;
        justify-content: center;
    }

    .hero h2 {
        font-size: 2.5em;
    }

    .features {
        flex-direction: column;
        align-items: center;
    }

    .feature-item {
        width: 90%;
    }
}
/* Mobile devices (smaller than 480px) */
@media (max-width: 480px) {
    header h1 {
        font-size: 1.8em;
    }

    header nav ul li {
        margin: 0 8px;
    }

    .hero {
        padding: 60px 15px;
    }

    .hero h2 {
        font-size: 2em;
    }

    .hero p {
        font-size: 1em;
    }

    .feature-item {
        padding: 15px;
    }

    footer .social-links {
        display: flex;
        justify-content: center;
        flex-wrap: wrap;
    }

    footer .social-links a {
        margin: 5px;
    }
}
```

OUTPUT:



CODE EXPLANATION:

HTML Structure

The webpage is divided into three main sections:

Header → Contains title and navigation links

Main → Displays content of the webpage

Footer → Shows copyright information

CSS Styling

Flexboxyling is applied for layout, colors, and spacing
Flexbox is used to align navigation links

TASK 2:

PROMPT:

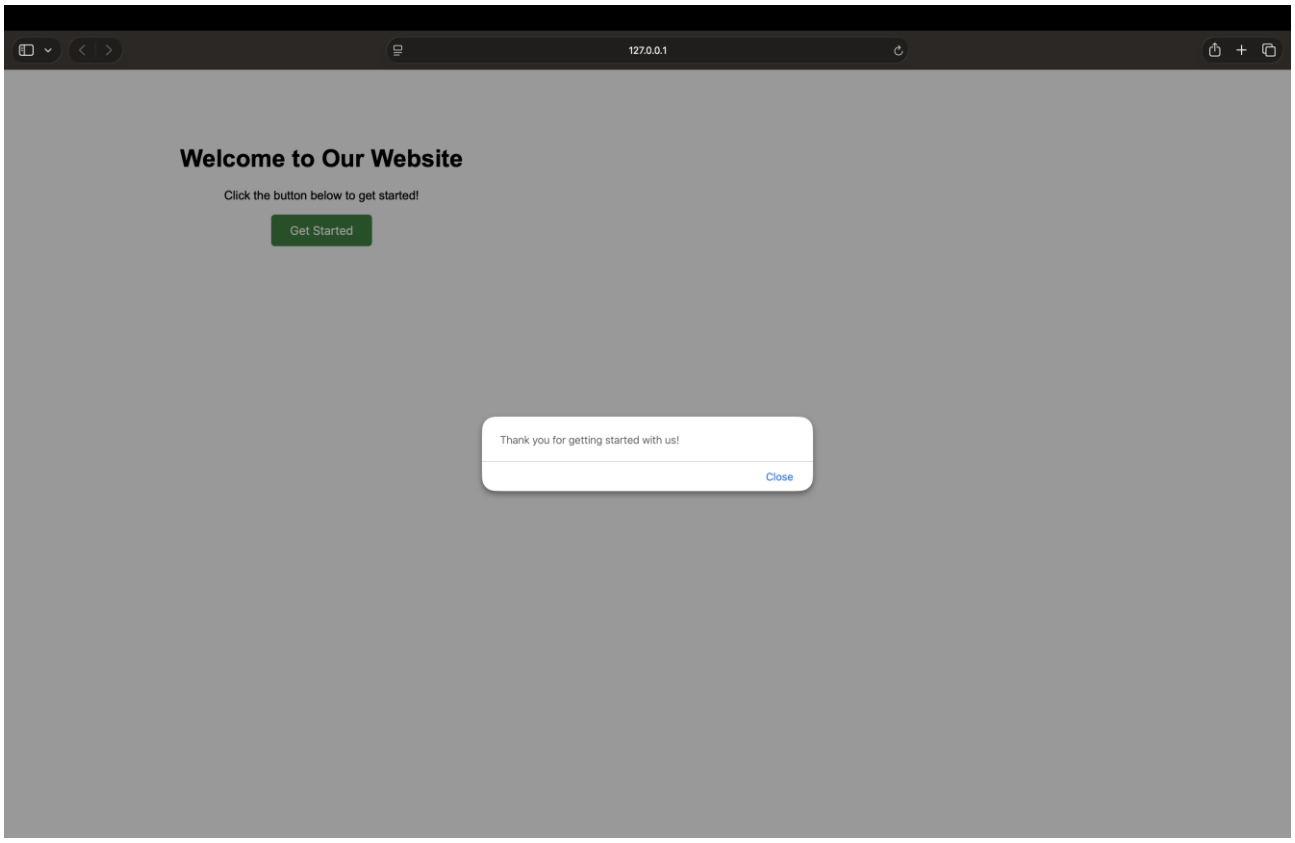
Generate HTML with a call-to-action button.
Add JavaScript that triggers an alert message when the button is clicked. Ensure the JavaScript code is well-commented and does not reload the page.

CODE:

TAST2.html > html > head > style > button:hover

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>CTA Button</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            text-align: center;
            margin-top: 100px;
        }

        button {
            padding: 12px 25px;
            font-size: 16px;
            background-color: #28a745;
            color: white;
            border: none;
            border-radius: 5px;
            cursor: pointer;
        }

        button:hover {
            background-color: #218838;
        }
    </style>
</head>
<body>

    <h1>Welcome to Our Website</h1>
    <p>Click the button below to get started!</p>

    <!-- Call-To-Action Button -->
    <button id="ctaButton">Get Started</button>

    <script>
        // Select the button using its ID
        const button = document.getElementById("ctaButton");

        // Add an event listener to detect button click
        button.addEventListener("click", function () {

            // This function runs when the button is clicked

            // Display an alert message to the user
            alert("Thank you for getting started with us!");

            // Note: No page reload happens because we are not submitting a form
        });
    </script>

</body>
</html>
```

OUTPUT:



CODE EXPLANATION:

HTML CODE:

A button element is ctaButtonsing <button>
It is given an id (ctaButton) to access it in JavaScript

JavaScript Logic:

document.getElementById() is used to select the button
addEventListener("click", function) is used to detect when the button is clicked
When clicked, an alert message is displayed

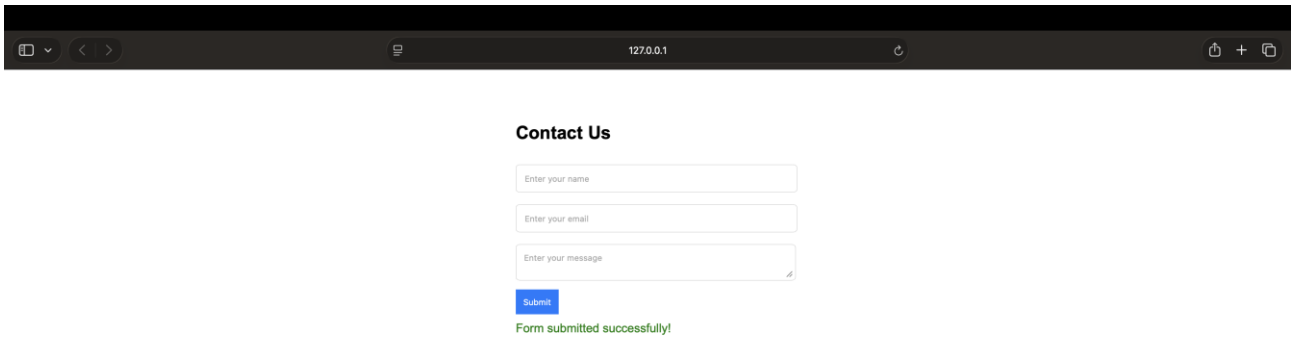
#TASK 3:
PROMPT:
Create an HTML contact form with fields for name, email, and message.
Add JavaScript validation to prevent empty inputs and check for valid email format.
Display inline error messages and show a success message when the form is valid.
Ensure validation happens on the client side without page reload.

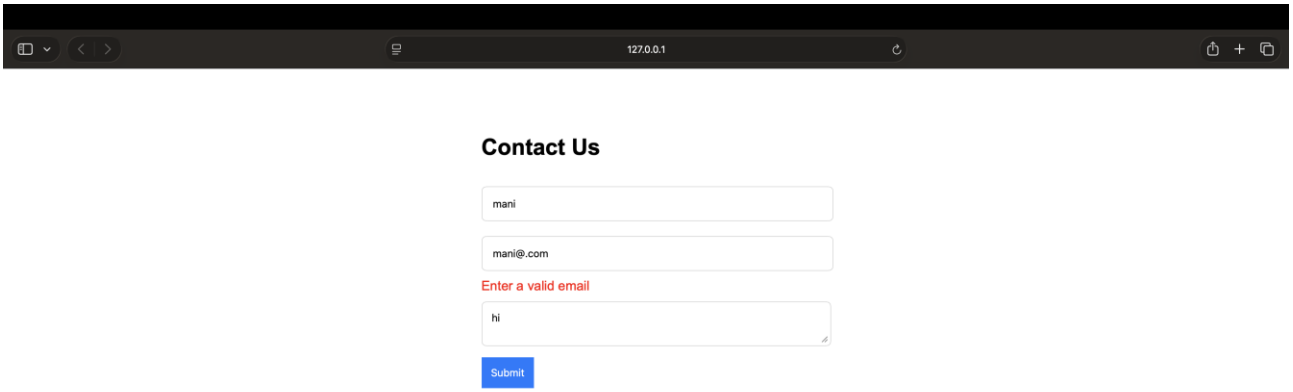
CODE:

```html
<html lang="en">
<body>
<div class="form-container">
    <h2>Contact Us</h2>

    <form id="contactForm">

        <!-- Name -->
        <input type="text" id="name" placeholder="Enter your name">
        <div id="nameError" class="error"></div>

        <!-- Email -->
        <input type="text" id="email" placeholder="Enter your email">
        <div id="emailError" class="error"></div>

        <!-- Message -->
        <textarea id="message" placeholder="Enter your message"></textarea>
        <div id="messageError" class="error"></div>

        <button type="submit">Submit</button>
    </form>

    <!-- Success Message -->
    <div id="successMsg" class="success"></div>
</div>
<script>
    // Get form element
    const form = document.getElementById("contactForm");

    // Add submit event
    form.addEventListener("submit", function(e) {

        // Prevent page reload
        e.preventDefault();

        // Get input values
        const name = document.getElementById("name").value.trim();
        const email = document.getElementById("email").value.trim();
        const message = document.getElementById("message").value.trim();

        // Get error elements
        const nameError = document.getElementById("nameError");
        const emailError = document.getElementById("emailError");
        const messageError = document.getElementById("messageError");
        const successMsg = document.getElementById("successMsg");

        // Clear previous errors
        nameError.textContent = "";
        emailError.textContent = "";
        messageError.textContent = "";
        successMsg.textContent = "";

        let isValid = true;

        // Name validation
        if (name === "") {
            nameError.textContent = "Name is required";
            isValid = false;
        }

        // Email validation
        const emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

        if (email === "") {
            emailError.textContent = "Email is required";
```

```
<script>
    form.addEventListener("submit", function(e) {

        if (email === "") {
            emailError.textContent = "Email is required";
            isValid = false;
        } else if (!email.match(emailPattern)) {
            emailError.textContent = "Enter a valid email";
            isValid = false;
        }

        // Message validation
        if (message === "") {
            messageError.textContent = "Message cannot be empty";
            isValid = false;
        }

        // If all valid
        if (isValid) {
            successMsg.textContent = "Form submitted successfully!";
            form.reset(); // Clear form
        }
    });
</script>
```

OUTPUT:
For Valid Submission:





For Invalid Submission:

CODE EXPLANATION:

Form Design

The form includes three input fields:
Name
Email
Message

Validation Logic

Name → must not be empty

Email → must follow correct format using regex

Message → must not be empty

Error Handling

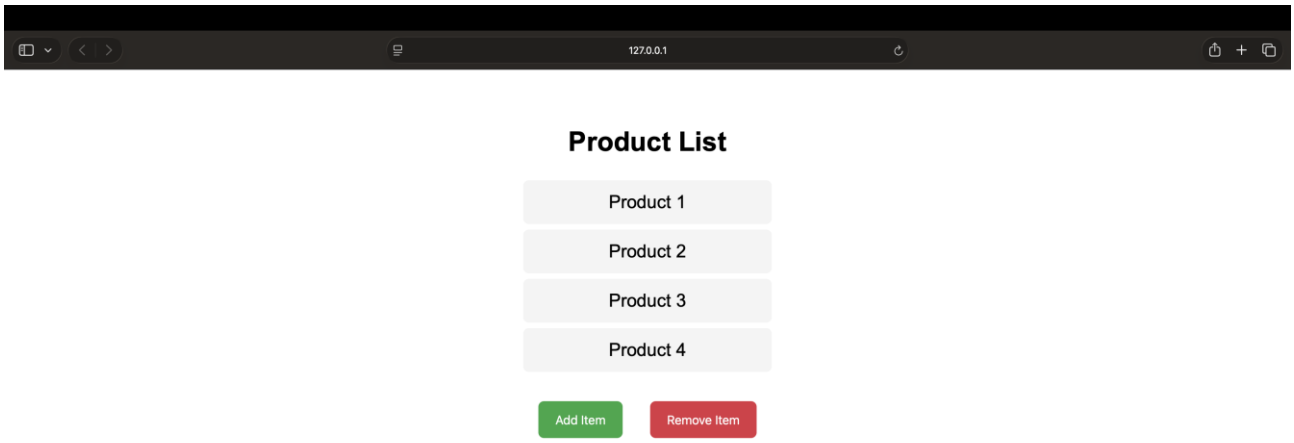
Error messages are displayed below each field
Previous errors are cleared before new validation

#TASK 4:
PROMPT:
Create an HTML page with a list of items. Add JavaScript functionality to dynamically add a new item when an "Add" button is clicked and remove the last item when a "Remove" button is clicked. Ensure the DOM updates instantly without page reload and include comments for clarity.

CODE:

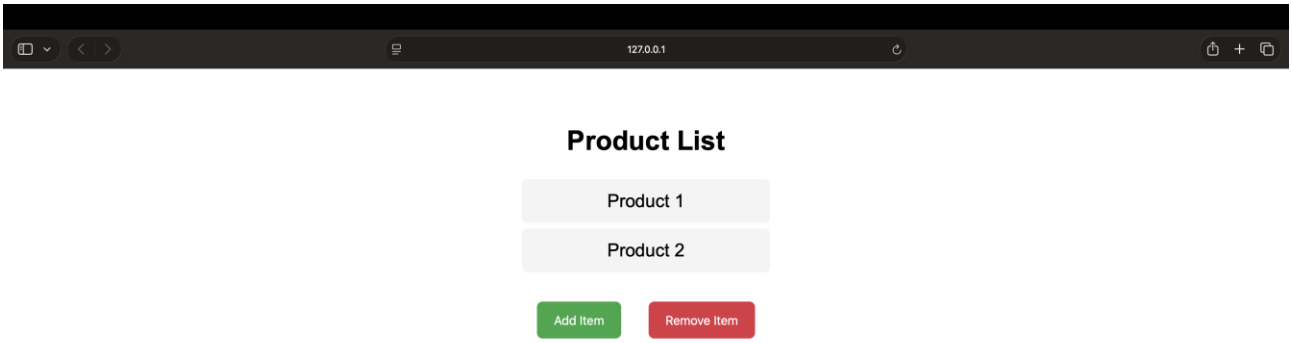
```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Dynamic List</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            text-align: center;
            margin-top: 50px;
        }

        ul {
            list-style: none;
            padding: 0;
        }

        li {
            background: #f4f4f4;
            margin: 5px auto;
            padding: 10px;
            width: 200px;
            border-radius: 5px;
        }

        button {
            margin: 10px;
            padding: 10px 15px;
            border: none;
            cursor: pointer;
            border-radius: 5px;
        }

        .add-btn {
            background: #28a745;
            color: white;
        }

        .remove-btn {
            background: #dc3545;
            color: white;
        }
    </style>
</head>
<body>
    <h2>Product List</h2>

    <!-- List -->
    <ul id="itemList">
        <li>Product 1</li>
        <li>Product 2</li>
    </ul>
    <!-- Buttons -->
    <button class="add-btn" onclick="addItem()">Add Item</button>
    <button class="remove-btn" onclick="removeItem()">Remove Item</button>

    <script>
        // Counter to keep track of item numbers
        let count = 3;

        // Function to add a new item
        function addItem() {
            // Select the list
            const list = document.getElementById("itemList");

            // Create a new list item
            const newItem = document.createElement("li");

            // Set text for new item
            newItem.textContent = "Product " + count;

            // Append item to list
            list.appendChild(newItem);

            // Increment counter
            count++;
        }
        // Function to remove last item
        function removeItem() {
            const list = document.getElementById("itemList");

            // Check if list has items
            if (list.children.length > 0) {
                // Remove last child
                list.removeChild(list.lastElementChild);
                count--;
            } else {
                alert("No items to remove!");
            }
        }
    </script>
</body>
</html>
```

OUTPUT:



Adds a new item when an "Add" button is clicked:

Removes the last item when a “Remove” button is clicked:



CODE EXPLANATION:

HTML Structure

An unordered list <ul> is used to display items
Two buttons:
Add Item
Remove Item

DOM Manipulation

The list updates instantly without page reload
JavaScript directly modifies the DOM (Document Object Model)

- **Created a dynamic product list**
- **Used JavaScript DOM manipulation**
- **Enabled real-time updates without refresh**
- **Implemented add/remove functionality**

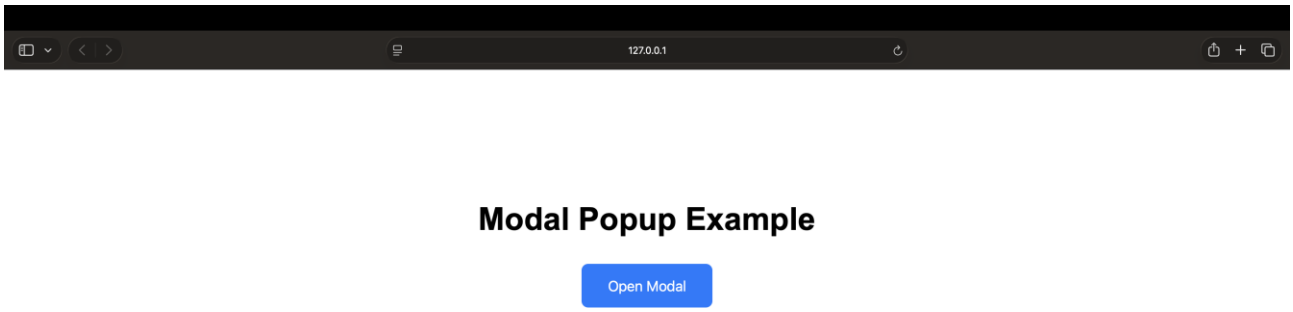
#TASK 5:
PROMPT:
Generate a modal popup using HTML, CSS, and JavaScript. Include a semi-transparent background overlay. Add functionality to open the modal using a button and close it using a close button or by clicking outside the modal. Ensure the code is clean and well-commented.

CODE:

```html
<html lang="en">
<body>

    <h2>Modal Popup Example</h2>

    <!-- Button to open modal -->
    <button class="open-btn" onclick="openModal()">Open Modal</button>

    <!-- Modal Structure -->
    <div id="modal" class="modal">

        <!-- Modal Content -->
        <div class="modal-content">

            <!-- Close Button -->
            <span class="close" onclick="closeModal()">&times;</span>

            <h3>Notification</h3>
            <p>This is a user notification popup!</p>

            <button onclick="closeModal()">Close</button>
        </div>
    </div>

    <script>
        // Select modal element
        const modal = document.getElementById("modal");

        // Function to open modal
        function openModal() {
            modal.style.display = "flex"; // Show modal
        }

        // Function to close modal
        function closeModal() {
            modal.style.display = "none"; // Hide modal
        }

        // Close modal when clicking outside content (overlay click)
        window.onclick = function(event) {
            if (event.target === modal) {
                closeModal();
            }
        };
    </script>

</body>
</html>
```

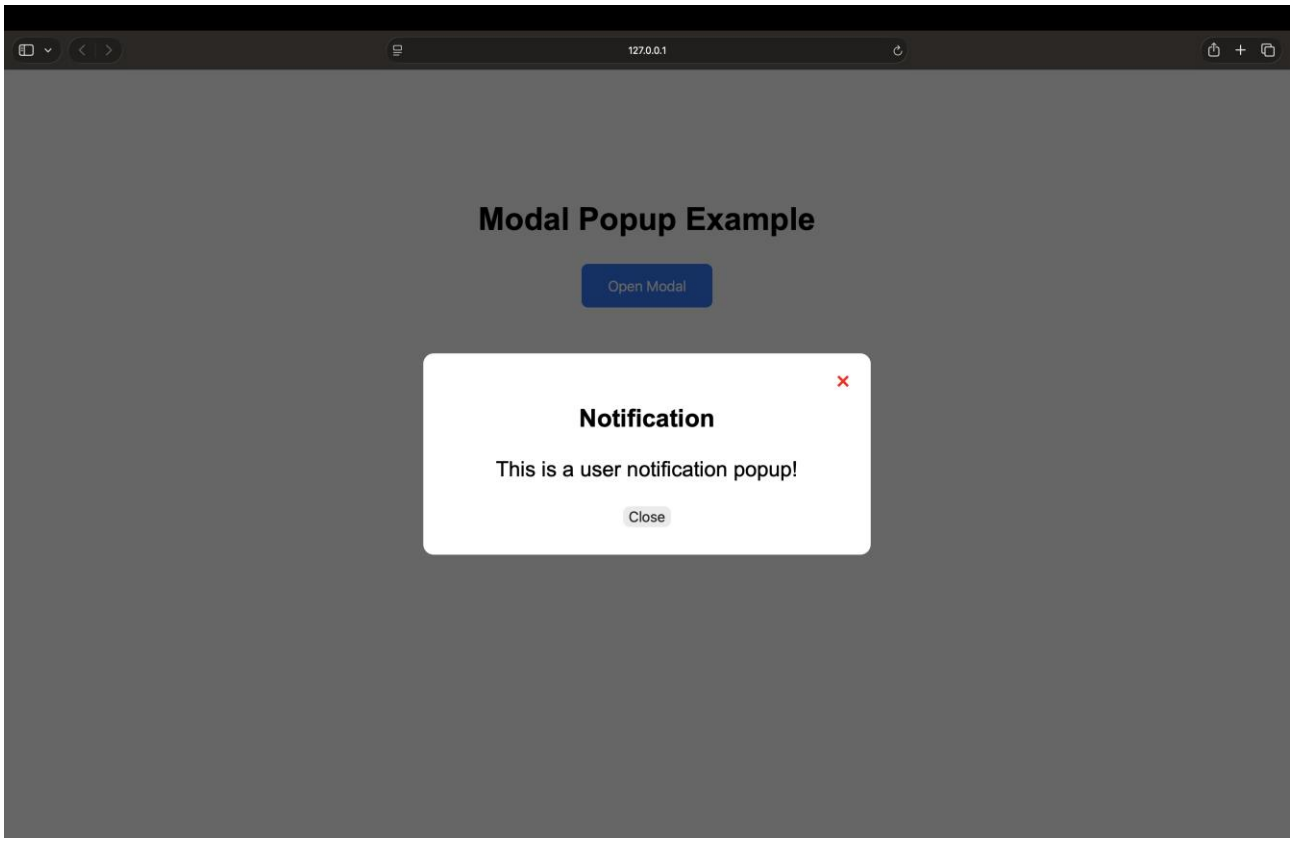
OUTPUT:

UserNotification Popup



CODE EXPLANATION:

HTML Structure



A button is used to open the modal
A with class modal acts as overlay
Inside it, modal-content contains popup message
A close button (×) is added

JavaScript Logic

➤Open Modal

Changes display from none → flex

➤Close Modal

Changes display from flex → none

➤Overlay Click

If user clicks outside modal content → modal closes

Features Implemented

- Open modal using button
- Close modal using:
- Close icon (×)
- Close button
- Clicking outside (overlay)